

Vehicle Currencies in AMMs: Model and Empirical Review Packet

Purpose: internal review before manuscript drafting.

Current status: no additional broad experiments are needed before writing if the paper is written around bounded claims. The remaining task is table hierarchy and claim discipline.

Bounded empirical spine:

1. Measurement: BridgeShare is conditional on indirect routing and must be paired with all-route bridge share.
2. P1: WETH vehicle routes provide availability and thin-direct-market protection in quoteable covered venues.
3. P2: vehicle-linked LP concentration and current bridge use predict future BridgeShare; this is not causal LP feedback.
4. P3: WETH downside shocks produce impact stress rotation toward stable vehicles; the supported window is same-day.
5. P4a: V3 evidence is the decline in no-direct/WETH-available cases, not broad V3 launch causality.
6. P4b: V4 partially separates route intermediation from physical intermediary-token transfer; size-bin evidence must be central.

Model Structure

Trader chooses between direct route $i \rightarrow j$ and vehicle route $i \rightarrow k \rightarrow j$.

$\text{DirectCost}(q) = f_D + s_D + \theta q / L_D$

$\text{VehicleCost}(q) = f_{IK} + f_{KJ} + s_K + \rho_K + \theta q (1/L_{IK} + 1/L_{KJ})$

$\text{RouteAdvantage} = \text{DirectCost} - \text{VehicleCost}$

$\text{BridgeShare}(\delta) = 1 / (1 + \exp(-\lambda \delta))$

P1 is not universal cost superiority. Vehicle route usefulness requires vehicle-route availability and one of: direct route unavailable, direct route thin, or vehicle route cheaper in common support.

P2 is reduced-form persistence:

$E[\text{BridgeShare}_{\{k,t+h\}}] = \alpha_k + \beta_L * \text{LPConcentration}_{\{k,t\}} + \rho * \text{BridgeShare}_{\{k,t\}}.$

P3 is interval-neutral in theory: vehicle risk lowers route advantage on impact. Same-day is the empirical implementation.

P4a increases direct-route liquidity, reducing relative vehicle reliance.

P4b netting intensity lowers physical intermediary-token movement while route intermediation can remain.

Mathematica Derivation Highlights

Key formal signs from output/model/model_derivations.txt:

P1AvailabilityProtection:

- $d\text{BridgeShare}/d\text{Advantage} = \lambda * \text{Sech}[\dots]^2 / 4 > 0$.
- VehicleUsefulCondition includes direct-unavailable, thin-direct, and positive-advantage cases.

P2LiquidityPersistence:

- $d\text{Advantage}/d\text{LIK} > 0$ and $d\text{Advantage}/d\text{LKJ} > 0$.
- $d\text{ExpectedBridgeShareNext}/d\text{VehicleLiquidity} = \beta_L \geq 0$.
- $d\text{ExpectedBridgeShareNext}/d\text{CurrentBridgeShare} = \rho \geq 0$.

P3StressRotation:

- $d\text{Advantage}/d\text{VehicleRisk} = -1$.
- $d\text{BridgeShare}/d\text{VehicleRisk} < 0$.

P4aConcentratedLiquidity:

- $d\text{Advantage}/d\text{DirectLiquidityMultiplier} < 0$.

P4bFlashAccounting:

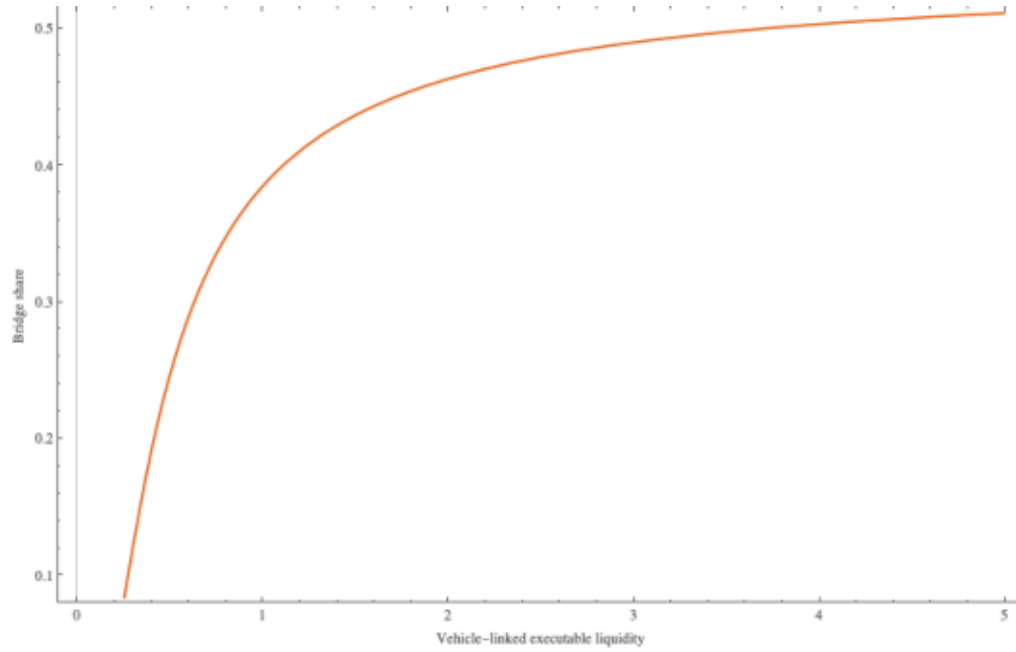
- $\text{PhysicalVehicleMovement} = (1 - n) * \text{GrossVehicleExposure}$.
- $\text{CompressionRatio} = n$.
- $d\text{PhysicalMovement}/d\text{Netting} < 0$; $d\text{Compression}/d\text{Netting} = 1$.

Full derivation object:

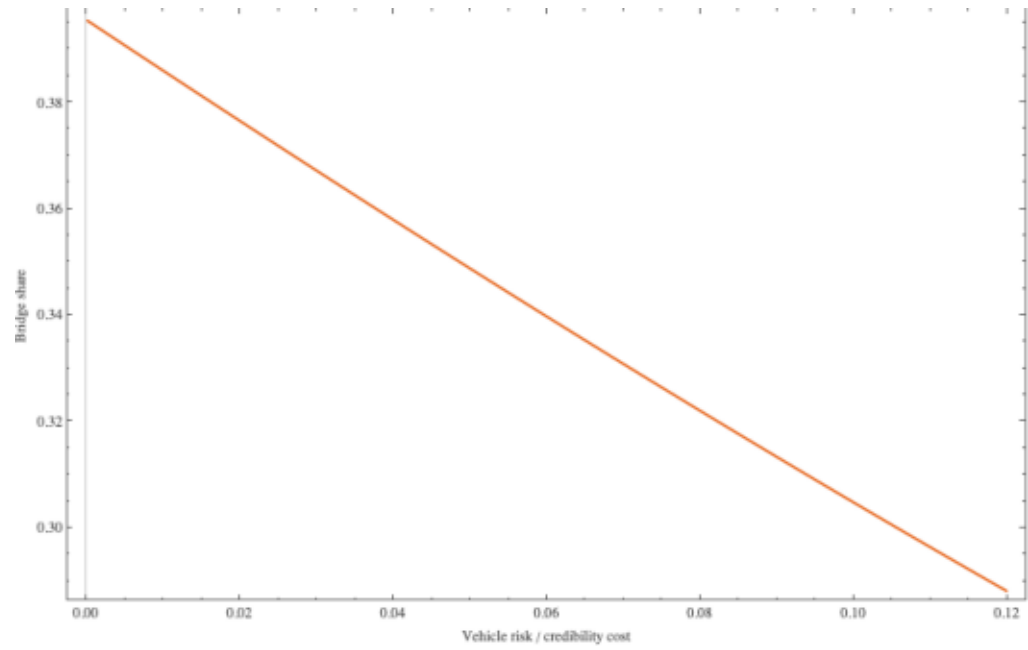
```
<|"P1AvailabilityProtection" -> <|"RouteAdvantage" -> fD - fIK - fKJ - rhoK + sD - sK + (q*theta)/LD - (LIK^(-1) + LKJ^(-1))*q*theta, "DirectAvailableCondition" -> LD >= Lmin, "VehicleRouteAvailableCondition" -> LIK >= Lmin && LKJ >= Lmin, "ThinDirectMarketCondition" -> LD <= Lthin, "VehicleUsefulCondition" -> LIK >= Lmin && LKJ >= Lmin && (LD < Lmin || LD <= Lthin || fD + sD + (q*theta)/LD > fIK + fKJ + rhoK + sK + (LIK^(-1) + LKJ^(-1))*q*theta), "dBridgeShare_dAdvantage" -> (lambda*Sech[(lambda*(fD - fIK - fKJ - rhoK + sD - sK + (LD^(-1) - LIK^(-1) - LKJ^(-1))*q*theta))/2]^2)/4|>, "P2LiquidityPersistence" -> <|"dAdvantage_dLIK" -> (q*theta)/LIK^2, "dAdvantage_dLKJ" -> (q*theta)/LKJ^2, "dBridgeShare_dLIK" -> (lambda*q*theta*Sech[(lambda*(fD - fIK - fKJ - rhoK + sD - sK + (LD^(-1) - LIK^(-1) - LKJ^(-1))*q*theta))/2]^2)/(4*LIK^2), "dBridgeShare_dLKJ" -> (lambda*q*theta*Sech[(lambda*(fD - fIK - fKJ - rhoK + sD - sK + (LD^(-1) - LIK^(-1) - LKJ^(-1))*q*theta))/2]^2)/(4*LKJ^2), "ExpectedBridgeShareNext" -> alphaK + betaL*Lv + r/(1 + E^(-(lambda*(fD - fIK - fKJ - rhoK + sD - sK + (q*theta)/LD - (LIK^(-1) + LKJ^(-1))*q*theta))))), "dExpectedBridgeShareNext_dVehicleLiquidity" -> betaL, "dExpectedBridgeShareNext_dCurrentBridgeShare" -> r|>, "P3StressRotation" -> <|"dAdvantage_dVehicleRisk" -> -1, "dBridgeShare_dVehicleRisk" -> -1/4*(lambda*Sech[(lambda*(fD - fIK - fKJ - rhoK + sD - sK + (LD^(-1) - LIK^(-1) - LKJ^(-1))*q*theta))/2]^2)|>, "P4aConcentratedLiquidity" -> <|"RouteAdvantageWithDirectLiquidityMultiplier" -> fD - fIK - fKJ - rhoK + sD - sK + (q*theta)/(a*LD) - (LIK^(-1) + LKJ^(-1))*q*theta, "dAdvantage_dDirectLiquidityMultiplier" -> -((q*theta)/(a^2*LD)), "dBridgeShare_dDirectLiquidityMultiplier" -> -1/4*(lambda*q*theta*Sech[(lambda*(fD - fIK - fKJ - rhoK + sD - sK + (1/(a*LD) - LIK^(-1) - LKJ^(-1))*q*theta))/2]^2)/(a^2*LD)|>, "P4bFlashAccounting" -> <|"GrossVehicleExposure" -> (LIK^(-1) + LKJ^(-1))*q, "PhysicalVehicleMovement" -> (LIK^(-1) + LKJ^(-1))*(1 - n)*q, "CompressionRatio" -> n, "dPhysicalMovement_dNetting" -> -((LIK^(-1) + LKJ^(-1))*q), "dCompression_dNetting" -> 1|>|>
```

Numerical Simulations

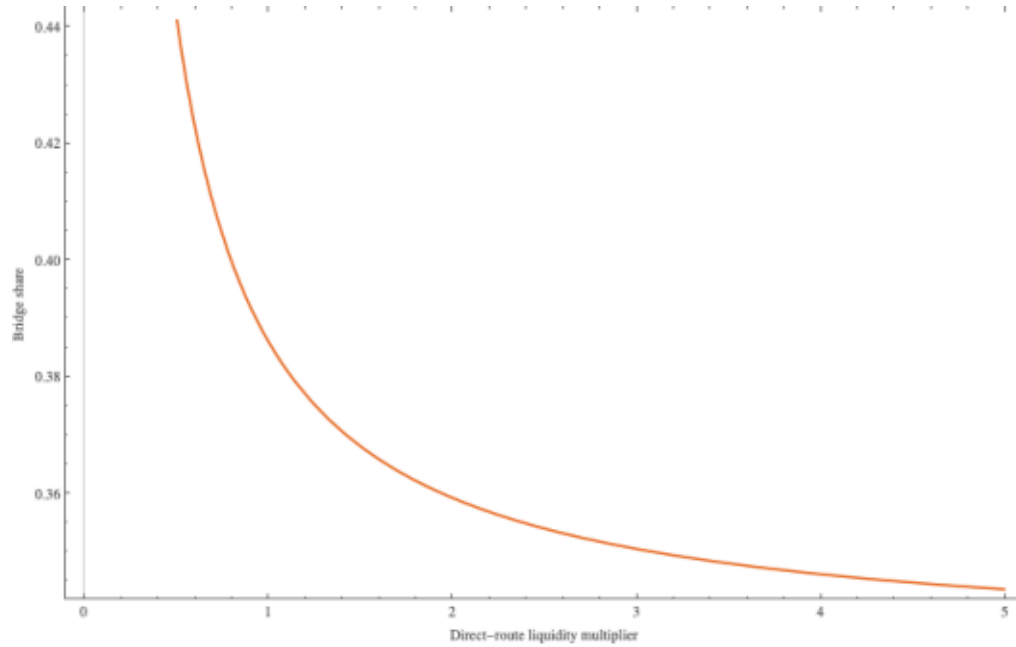
bridge_share_liquidity



bridge_share_risk



bridge_share_direct_liquidity



v4_netting_compression

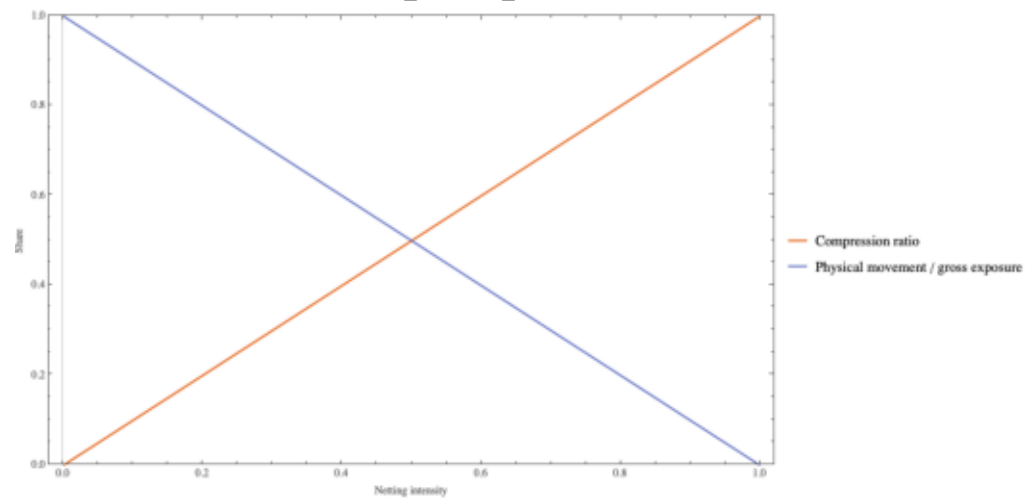


Table 1. Measurement and Scope

Panel		Quantity	Main estimate	Scope / companion estimate	
A. Vehicle-use denominators, 2026		WETH	44.5% conditional BridgeShare	5.3%	all-route bridge share
50.0% endpoint-pair coverage					
A. Vehicle-use denominators, 2026		USDC	20.6% conditional BridgeShare	2.5%	all-route bridge share
25.3% endpoint-pair coverage					
A. Vehicle-use denominators, 2026		USDT	23.6% conditional BridgeShare	3.0%	all-route bridge share
20.7% endpoint-pair coverage					
B. Exact-quote scope	Exact-quote covered share		78.9	NaN	Unified leg-volume share covered by V2/Sushi V2/V3 plus Bal
B. Exact-quote scope	Excluded Curve+Fluid share		16.2	NaN	Unified leg-volume share not covered by
B. Exact-quote scope	Excluded / covered ratio		0.205	NaN	Maximum exact-quote scope exposure relativ
B. Exact-quote scope	Excluded stable-leg share		75.3	NaN	Excluded venues are stablecoin-heavy, so ro

Table 2. P1 Availability and Thin-Direct Protection

Trade size	Direct available (%)	WETH route available (%)	No-direct / WETH-available rows	Thin-direct median advantage (bp)	
High-quality-direct	median advantage (bp)	Common-support	median (bp)		
\$1,000	72.1	67.4	9,584	142.65	8.95
\$10,000	72.1	67.4	9,584	190.21	
-1.12	21.52				
\$100,000	72.1	67.4	9,584	349.28	
-76.16	-0.39				

Table 3. P2 Dynamic Predictability

Horizon	Baseline future BridgeShare (%)	LP beta	LP SE	LP p	Effect of +10pp LP concentration (pp)	Persistence beta		
Persistence SE	Persistence p	Effect of +10pp current BridgeShare (pp)						
t+1		18.27	0.120	0.012	<0.001	1.20	0.586	0.019
t+7		18.26	0.142	0.013	<0.001	1.42	0.460	<0.001
t+14		18.26	0.140	0.014	<0.001	1.40	0.404	
0.018	<0.001				4.04			
t+30		18.24	0.148	0.013	<0.001	1.48	0.330	
0.016	<0.001				3.30			

Table 4. P3 Impact Stress Rotation

Panel	Estimate	Events	Effect	t	p	Interpretation
A. Main decomposed same-day effect prior-28-day baseline	WETH share change	20	-1.48 pp	-2.59	0.018	event-day minus
A. Main decomposed same-day effect prior-28-day baseline	Stable share change	20	1.48 pp	2.59	0.018	event-day minus
A. Main decomposed same-day effect prior-28-day baseline	WETH-minus-stable change	20	-2.96 pp	-2.59	0.018	event-day minus
A. Main decomposed same-day effect prior-28-day baseline	Aggregate direct-route share change	20	-0.25 pp	-0.42	0.679	event-day minus
A. Main decomposed same-day effect prior-28-day baseline	Log indirect-route volume change	20	0.51 log points	3.20	0.005	event-day minus
B. Threshold/overlap sensitivity negative share 67.0%	all events, threshold 6%	91	-3.12 pp	-3.44	<0.001	
B. Threshold/overlap sensitivity negative share 65.3%	all events, threshold 8%	49	-2.95 pp	-3.41	0.001	
B. Threshold/overlap sensitivity negative share 65.4%	all events, threshold 10%	26	-3.74 pp	-3.70	0.001	
B. Threshold/overlap sensitivity negative share 63.2%	all events, threshold 12%	19	-3.18 pp	-2.68	0.015	
B. Threshold/overlap sensitivity negative share 61.1%	top 20, threshold 8%	18	-3.09 pp	-2.48	0.024	
B. Threshold/overlap sensitivity negative share 57.6%	non-overlap, threshold 8%	33	-3.01 pp	-2.45	0.020	

Table 5. P4a V3 Route Opportunity

	Outcome	Rows	Pairs	Post-V3	effect	t	p	Pretrend	slope	Pretrend t	Pretrend p	Units
No-direct WETH	availability	1,822	190		-25.81	-7.62	<0.001		0.071	0.1	0.922	pp

Table 6. P4b V4 Settlement Virtualization

	Panel Sample / diagnostic	Cells	V3	V4	Difference / balance	p
A. Transfer incidence by route-size bin	All	500	100.0%	81.4%	-18.6 pp	<0.001
A. Transfer incidence by route-size bin	Route size: Small	104	100.0%	60.6%	-39.4 pp	<0.001
A. Transfer incidence by route-size bin	Route size: Medium	73	100.0%	87.7%	-12.3 pp	0.002
A. Transfer incidence by route-size bin	Route size: Large	94	100.0%	96.8%	-3.2 pp	0.083
B. Matched-sample balance	uniswap_v3	500	\$6,136	NaN	53.05	100.0
B. Matched-sample balance	uniswap_v4	500	NaN	\$3,542	71.05	81.4
B. Matched-sample balance	V4 - V3 within cell	500	NaN	NaN	log route diff=-0.692 t=-7.58, p=<0.001	

Table 7. Specification Registry

Test	Unit	Sample	Outcome	Regressor / treatment
P1 availability/thin-direct endpoint-pair x day x trade size V2/Sushi V2/V3 exact quoteable venues route availability and WETH advantage	WETH vehicle route endpoint-pair-day aggregation	direct available 72.1% 9,584		
no-direct rows; thin-direct 142.65-349.28 bp	availability and thin-direct execution protection			
P2 predictability	token x day	vehicle candidates	future BridgeShare LP concentration; current BridgeShare tok	
P3 impact stress event x common endpoint-pair set	WETH downside event days	WETH-minus-stable BridgeShare	event-day WETH downside stress	
P4a V3 opportunity	endpoint-pair x month	balanced V3 launch-window pairs	no-direct/WETH-available indicator	post-V3
P4b V4 settlement	matched route unit	matched V3/V4 cells intermediary-token transfer incidence		V4 route unit